The Role of Water in the Superconductivity of Sodium Cobaltate

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Sodium cobaltate superconducts only when it is wet, and after twenty years there is no agreed reason why. We take the water seriously as a mechanical object and calculate, by ordinary density-functional methods, what happens between the CoO_2 sheets when the gallery that holds the sodium is pulled open. Past a critical spacing two things happen at once: the charge on the oxygen layers bifurcates, so an electron gas condenses in the gallery, and the stiff sodium well splits into a shallow double well in which the ion begins to rattle. The gas supplies carriers and the rattle supplies glue, and a phonon-mediated superconducting dome sits in the narrow window between the layers too close, where there is no gas, and the sodium frozen off-centre, where the coupling runs away into a polaron. The dry parent and the monolayer hydrate fall correctly outside this window. One number does not fit: the superconducting bilayer hydrate holds its layers wider than the static window, and resolving that mismatch forces the paper's one real claim about the water. The water is not a spacer. It is a lubricant, and that statement can be tested.

I. INTRODUCTION

It is a curious fact that sodium cobaltate becomes a superconductor only when it is wet. Dry Na_xCoO_2 is an ordinary metal down to the lowest temperatures measured; slip two layers of water between its CoO_2 sheets and it superconducts near 4.5 K [1]; and twenty years of looking for the reason inside the CoO_2 sheet itself have not produced an accepted mechanism [2]. This paper takes the water seriously as a mechanical object and asks what it does. We shall show, by ordinary density-functional calculations, that pulling the layers apart past a critical separation does two things at once: the charge on the oxygen layers bifurcates, so that an electron gas condenses in the gallery between the sheets, and the sodium ion, which had been sitting at the bottom of one stiff well, finds itself with two shallow wells and begins to rattle between them. The gas supplies the carriers and the rattle supplies the glue, and superconductivity can live only in the window between the layers too close, where there is no gas, and the sodium frozen, where there is no glue (Fig. 1).

There is nothing exotic in the calculations, and the ingredients—a two-dimensional electron gas, an anharmonic mode—are old ones; what is new is only where we look for them, which is between the layers rather than in them. The electronic background is the one worked out by Radha and Lambrecht for the surfaces of the layered cobaltates [3], whose surface bands and their spin polarization are the seed of everything below; the present paper is its sequel, moved from the surface into the gallery.

One measured number does not fit this static picture. The superconducting bilayer hydrate holds its layers 9.9 Å apart, wider than the window we calculate, and there the static calculation says the sodium is not rattling at all but bound hard against one CoO_2 layer. We shall meet this difficulty early, in Sec. IV, because resolving it is what teaches us the actual role of the water. The water is not a spacer. It is a lubricant.

We first show what the calculations find when the gallery opens, told as a single event seen three ways. We then build the electron–phonon model that turns that event into a superconducting dome and locate its two walls. We take up the 9.9 Å difficulty in its own section. Finally we ask what the picture must then explain—why lithium never superconducts, where the magnetic phase sits, what a valence probe should read—and where the calculation is weakest.

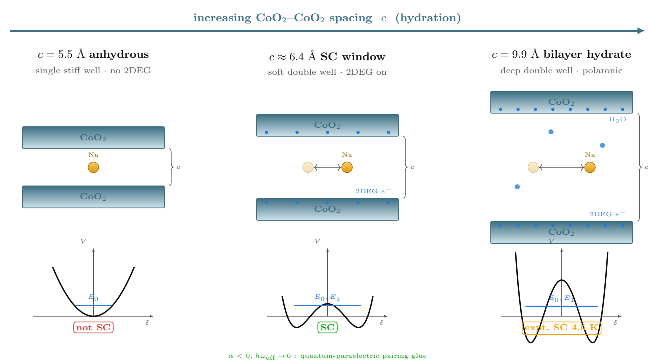

FIG. 1. The mechanism in one picture. When the gallery between the CoO_2 sheets is narrow (left), the sodium sits at the centre of a single stiff well and the electrons stay bound to the oxygen: no gas, no soft mode, no pairing. Open the gallery past a critical spacing (right) and both change at once—charge spills back into the gallery as a two-dimensional electron gas, and the sodium well splits into a shallow double well in which the ion rattles. The gas carries and the rattle glues. Superconductivity lives in the narrow window between these two states and the polaronic state past it.

II. WHAT HAPPENS WHEN THE GALLERY OPENS

Picture the sodium ion sitting in the gallery, midway between two CoO_2 sheets, and slowly pull the sheets apart. While the gallery is narrow the ion sits at the centre in a single stiff well; there is one lowest-energy place for it and it stays there. Open the gallery far enough and the centre stops being the bottom of the well and becomes the top of a small barrier: the ion now has two equivalent places to sit, one toward each sheet, and between them it rattles. At the same spacing, and this is the point, the electrons do something of their own. The charge that the oxygen layers had been holding splits, and part of it spills back into the gallery as a two-dimensional electron gas. One event, two faces.

We see this in the total energy directly. For each spacing c we move the sodium off centre by a displacement δ and record the energy $E(\delta)$; the calculations are spin-polarized PBE with projector-augmented waves, at fixed in-plane geometry, and are described in Sec. VI. Near the bottom the curve is even in δ, so it is fit by

$$E(\delta) = E_0 + \alpha\,\delta^2 + \beta\,\delta^4, \tag{1}$$

and the sign of the quadratic coefficient α is the whole story: while $\alpha > 0$ the centre is stable and the ion has one well; when $\alpha < 0$ the centre is unstable and the well has split in two. Figure 2 shows the crossover. At $c = 5.5$ Å, the dry spacing, $\alpha = +2.5$ eV/Å^2 and the well is single and stiff. By $c = 6.9$ Å, the monolayer-hydrate spacing, $\alpha = -0.5$ eV/Å^2 and the ion has a pair of wells about 0.7 Å off centre. The sign flips between two spacings we actually computed, so the critical spacing is bracketed by the data, 5.5 Å $< c^*_{\mathrm{Na}} <$ 6.9 Å; interpolating the four points puts it at $c^*_{\mathrm{Na}} \approx 6.40$ Å. Repeating the whole scan at the physical composition $x = 1/3$, in a $\sqrt{3}\times\sqrt{3}$ cell, moves the crossing by only about 0.2 Å. The transition is not an artifact of the full-coverage cell.

The electron gas turns on at the same place, and we can watch it two independent ways. The first is the density of states at the Fermi level, $N(0)$, computed from a dense $\mathbf{k}$ mesh: for the Na 1×1 series it is 0.03 states per eV per cell per spin at 5.5 Å and jumps to 1.06 at 6.9 Å, a factor of thirty, and then saturates. Below the transition the gallery is insulating; above it there are carriers. The second is the Bader charge that flows back onto the sodium as the gallery opens, $\Delta q = 0/0.14/0.40/0.58\ e$ across 5.5/6.9/8.4/9.9 Å; read as an areal density this is a few times 10^{13} carriers per cm^2, the scale of a real two-dimensional electron gas. The two probes are measuring different things—one the band edge crossing E_F, the other the occupation filling up—and they agree on when the gas appears. The soft mode and the gas appear together because they are one event: opening the gallery both unbinds the electrons from the oxygen and unbinds the sodium from the centre.

Figure 3 collects the three signatures on one axis, and it is the figure the rest of the paper points back at: the Landau coefficient $\alpha(c)$ crossing zero, the carrier density $N(0)$ switching on, and—built on top of them—the superconducting dome we come to next. Two warnings belong here, not later. First, beyond about 7 Å the "double well" is no longer really a double well: the energy is still falling at the largest displacement we sampled, which

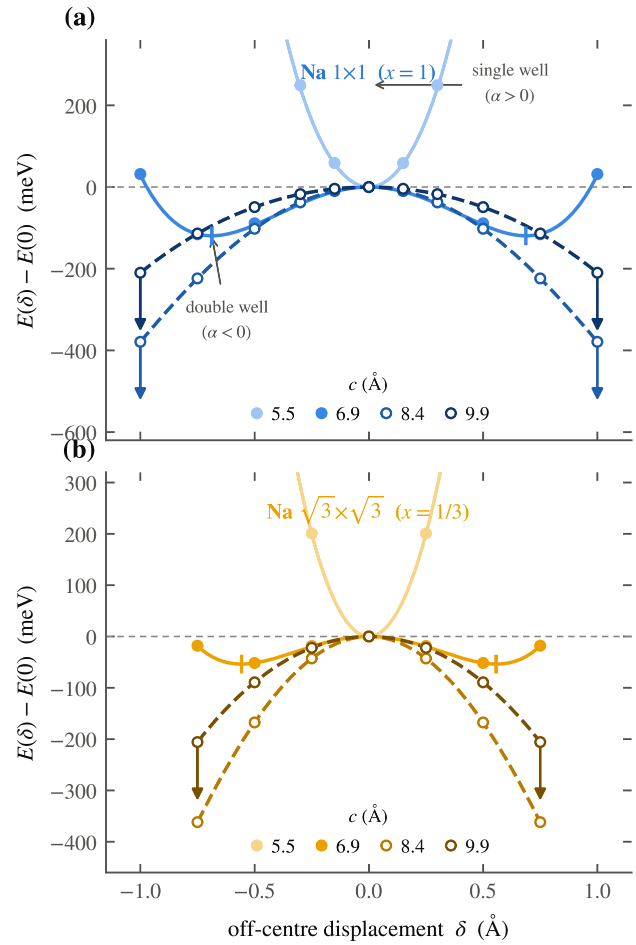

FIG. 2. The single-to-double-well transition. Energy $E(\delta)$ of the off-centre sodium displacement, computed for Na 1×1 (top, $x = 1$) and Na $\sqrt{3}\times\sqrt{3}$ (bottom, $x = 1/3$) at the four gallery spacings; points are total energies, curves are the fits of Eq. (1) continued to sixth order where needed. At 5.5 Å the well is single and stiff ($\alpha > 0$); by 6.9 Å it has split into a double well ($\alpha < 0$) with minima marked. Open symbols with arrows ($c \geq 8.4$ Å) flag scans that are still falling at the largest sampled displacement—there the sodium is adsorbing onto one layer and the quoted depth is only a lower bound.

means the sodium is sliding toward a bound site on one CoO_2 layer and the calculation has not reached the minimum. Those depths are lower bounds, and we mark them as such. Second, the well depths themselves carry a large systematic, discussed in Sec. VI, because we held the oxygen planes fixed. The sign of α does not; the single-to-double-well flip survives every check we ran.

III. THE PAIRING WINDOW

Now we make the mechanism quantitative, and it is worth saying first what is borrowed and what is done here. The band structure is borrowed: the SSH-

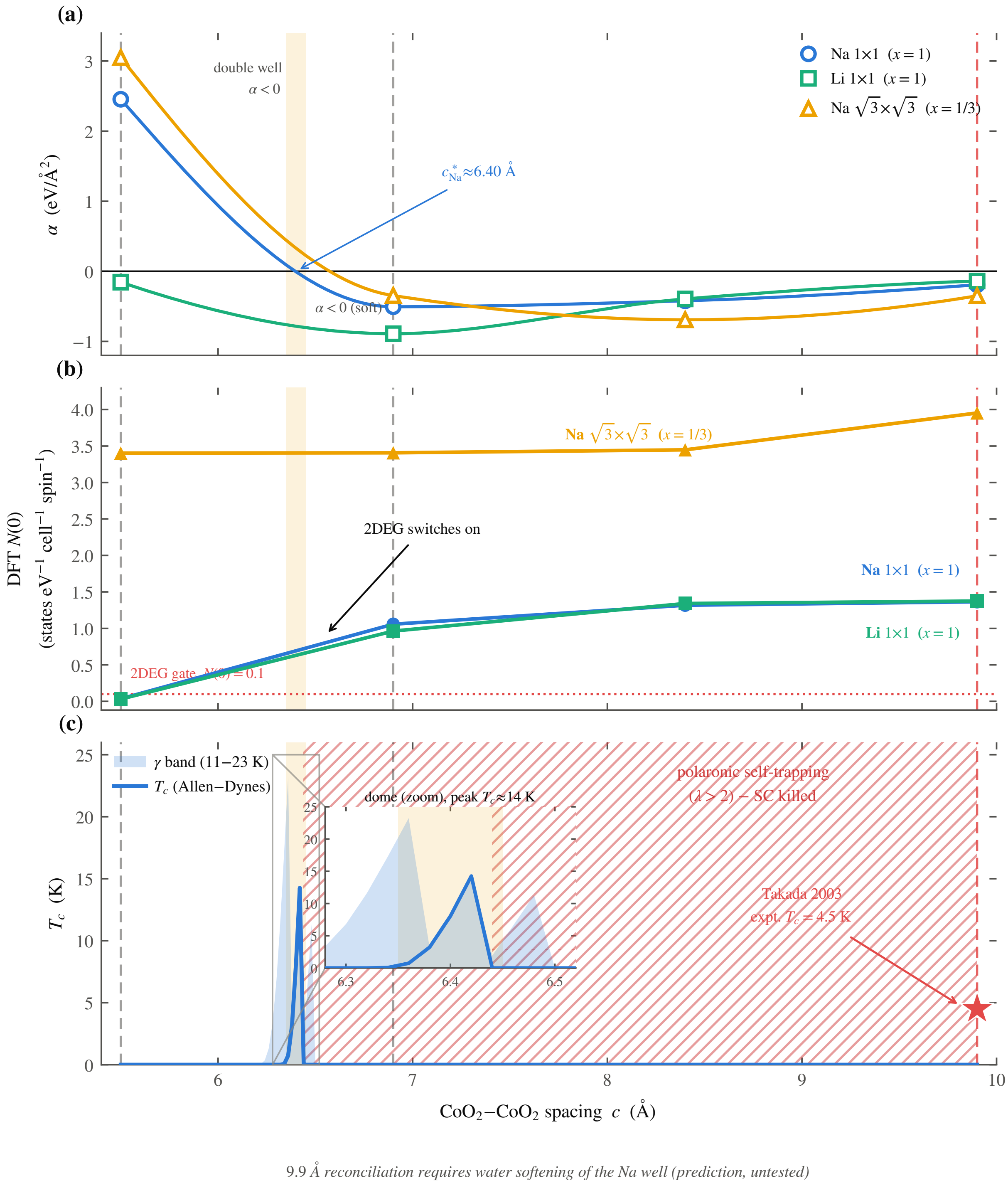

FIG. 3. The central result, three signatures of one transition against the CoO_2–CoO_2 spacing c. (a) The Landau coefficient $\alpha(c)$ for the three series; it crosses zero near $c^*_{\mathrm{Na}} \approx 6.40$ Å, where the single well becomes a double well. (b) The Fermi-level density of states $N(0)$, which jumps from essentially zero to of order one state per eV at the same spacing—the interlayer electron gas switching on. (c) The Allen–Dynes superconducting temperature: a narrow dome pinned just above c^*, with the shaded band spanning the electron–phonon coupling ansatz (11–23 K) and the hatched region where the coupling exceeds the polaron limit and superconductivity is killed. The experimental anchors are marked: the dry parent and monolayer hydrate (not superconducting) fall on the walls, and the bilayer hydrate (star, $T_c = 4.5$ K) does not sit in the static window—the difficulty of Sec. IV.

type four-band tight-binding model of Radha and Lambrecht [3] describes the two surface bands—one of alkali sp_z character sitting in the gallery, one of CoO_2 character—that share a single electron according to how their centres are split. The splitting is set by the Bader charge we computed, so the gallery band fills exactly as the gas turns on in Fig. 3(b). What is done here is to let the sodium move and ask how the electrons feel it.

The coupling has a parity to it that decides the form of the answer. The alkali displacement δ is odd under reflecting the gallery through its midplane, so a linear electron–phonon coupling, which would shift the band energy in proportion to δ, must vanish identically at the symmetric point. We checked this in the model at several k_z and it holds. The leading coupling is therefore quadratic: the band energy responds as $\frac{1}{2}\chi\delta^2$, with

$$\chi = 2\sum_m \frac{|\langle \mathrm{alk}|\,\partial H/\partial\delta\,|m\rangle|^2}{E_{\mathrm{alk}} - E_m}, \tag{2}$$

the ordinary second-order perturbation sum, which comes out to $\chi \approx -6$ eV/Å^2 near the transition and grows as the band splitting collapses at wider spacing. Because the coupling is quadratic, an electron does not scatter off one quantum of the rattling mode; it scatters off two. The relevant matrix element is $\langle 0|\delta^2|2\rangle$ between the ground state and the second excited state of the quantized well—the first excited state is forbidden by the same parity—and the effective boson is the even overtone $\hbar\Omega = E_2 - E_0$. The dimensionless coupling is then

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \qquad g = \tfrac{1}{2}|\chi|\,\langle 0|\delta^2|2\rangle, \tag{3}$$

with $N(0)$ taken from the density-functional calculation, not fit.

The window has two walls, and each is a sentence. Close the gallery below c^* and $N(0) \to 0$: there are no carriers, so $\lambda \to 0$ and there is nothing to pair. Open it too far, or cool the sodium into one well, and the ion freezes off-centre while $\hbar\Omega$ stays near 20 meV; the coupling then runs up past $\lambda \approx 20$, which is not strong superconductivity but a self-trapped polaron, and the pairing is killed. Between the two the quantized well is soft: the effective zero-point frequency of the sodium mode collapses from $\hbar\omega_{\mathrm{eff}} = 30$ meV in the stiff single well to a fraction of a meV in the double well (Fig. 4), which is the soft, strongly anharmonic mode the mechanism needs. Feeding $\lambda(c)$, $\hbar\Omega(c)$ and $N(0)$ into the Allen–Dynes formula, with $\mu^* = 0.10$ and gated on $N(0) \geq 0.1$, gives the dome of Fig. 3(c): a narrow peak between $c \approx 6.35$ and 6.45 Å, of height $T_c \approx 14$ K at $\lambda \approx 1.3$ and $\hbar\Omega \approx 11$ meV. The dry parent (5.5 Å) sits on the low wall with no gas, and the monolayer hydrate (6.9 Å) sits past the high wall in the polaronic regime; both come out correctly non-superconducting.

We state the honest status of the number 14 K. The height of the dome depends on the electron–phonon range parameter as the fourth power, and over its plausible range the peak runs from about 11 to 23 K; the width, near 0.1 Å, depends on interpolating the well shape between four computed spacings and is not a precision statement. What is robust is qualitative and is the claim of this section: a dome exists, it sits pinned just above the gas turn-on and below the polaron crossover, its height is of the right order for a few-kelvin superconductor, and it contains neither the monolayer nor the bilayer anchor. The last of those is a problem, and we turn to it now.

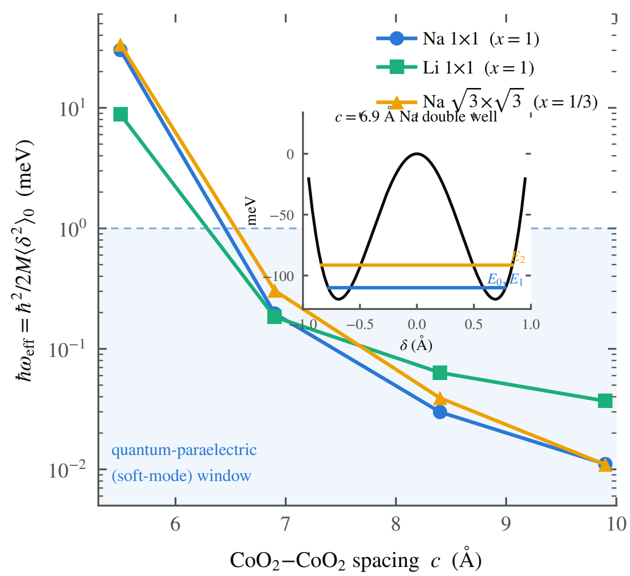

FIG. 4. The sodium mode softening. Effective zero-point frequency $\hbar\omega_{\mathrm{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, from an exact one-dimensional solution of the fitted wells at the physical masses. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode into a sub-meV soft mode as the gallery opens, entering a quantum-paraelectric window. Inset: the lowest levels in the 6.9 Å sodium double well, with the near-degenerate tunnelling doublet E_0, E_1 and the even overtone E_2 that carries the quadratic electron–phonon coupling.

IV. THE DIFFICULTY AT 9.9 Å AND WHAT IT TEACHES

The bilayer hydrate is the compound that actually superconducts, and it holds its CoO_2 layers 9.9 Å apart [4]. That is far outside the dome we calculated. Worse, when we compute the sodium well at 9.9 Å without water, we do not find a soft symmetric rattler at all: the energy falls all the way to the largest displacement we sampled, meaning the sodium is not centred but bound hard against one CoO_2 sheet. The static picture puts the one superconducting compound in the dead polaronic regime. A lesser paper would leave this in a supplement. It is the most important number here, and it has, we believe, a single cause: the calculation holds the water still, and the water does not hold still.

Consider what is actually in that gallery. At 9.9 Å the water bilayer fills nearly four-fifths of the space between the sheets, and it is a hydrogen-bonded network, not a rigid spacer. A frozen calculation sees only the average positions and reports a deep asymmetric well. But a mobile, polarizable water network screens the attraction that would otherwise bind the sodium to one layer, and its fluctuating hydrogen bonds shake the ion back and forth across the midplane. The effect on the sodium potential is to shallow it and re-symmetrize it—to push the well back toward the soft, near-critical shape that lives in the dome. The water is not what sets the layers 9.9 Å apart and then stands aside. The water is what keeps the sodium soft at a spacing where, left to a static lattice, it would freeze. It is a lubricant, not a spacer.

This converts the mismatch into the sharpest prediction in the paper, and it is a prediction about dynamics, so the tests are about dynamics. First, deuteration: replacing H_2O by D_2O stiffens the water network's librations by roughly $\sqrt{2}$ and tightens its hydrogen bonds, which should shift the sodium well and therefore T_c—a small, possibly inverse, isotope effect on T_c with the exponent well below the BCS value of one half, because the boson is the sodium and not the water. Second, spectroscopy: there must be a soft, strongly anharmonic sodium c-axis mode at the meV scale in the superconducting hydrate, visible to inelastic neutron scattering or low-frequency Raman, and it must be dynamical—sodium seen statically split, sitting off-centre without rattling, would kill the mechanism. Third, and decisive, the calculation we have not done: $E(\delta)$ for the sodium between explicit water layers. If that calculation still shows the sodium bound hard to one CoO_2 sheet, the water does not do what we say and the picture is wrong. This is the next calculation for this problem, and we say plainly that the mechanism stands or falls on it.

V. CONSEQUENCES

A mechanism is worth only what it forbids and predicts. Figure 6 lays the picture against the experimental record fact by fact; here we take the consequences one at a time.

a. Why lithium never superconducts. Lithium cobaltate is the same structure with a smaller ion, and it has never been made to superconduct. Our picture requires it. Two things go against lithium at once. Its well crosses into the double-well regime only at the smallest spacing in our dataset, so we have only an upper bound $c^*_{\mathrm{Li}} \leq 5.5$ Å, and at that spacing the well is a mere 4 meV deep while the lithium zero-point energy is 8.8 meV—larger than the well itself. The light ion does not sit in either minimum; it is smeared across both. This is a quantum paraelectric, dynamically centred despite the negative α, and a centred ion is a stiff ion with no soft anharmonic mode to offer. Lithium fails on the glue, and it fails for the plain reason that it is light.

b. The magnetic strip and the NMR contradiction. Right where the gas first condenses, the gallery band is narrow and its density of states is high, and a narrow band with a high density of states is a Stoner magnet. The Radha–Lambrecht surface calculation already finds this polarized gallery gas [3], with the gallery moment antiparallel to the cobalt. In our picture the magnetism is therefore a thin strip hugging the carrier turn-on, strongest just past it and dying as the band widens—so superconductivity lives one step beyond the magnetism, and the magnetism lives one step beyond the turn-on. This is the ordering Sakurai, Ihara and Takada report for the magnetic phase bordering the dome against cobalt valence [5]. It also dissolves an old contradiction. Early powder NMR suggested a triplet, later single-crystal work a singlet; in this picture that is not a contradiction but a statement of where each sample sits, because the Stoner enhancement varies steeply across the strip and hydration drifts from sample to sample. The prediction is sharp: the NMR Knight-shift suppression below T_c should anticorrelate, across samples, with the normal-state Stoner enhancement. Both symmetry-allowed states of Mazin and Johannes [2] are consistent with the muon null [6], and we do not need the chiral $d + id$ scenario [7] to fit the record, though our data do not exclude it.

c. Spin–orbit coupling. One might worry that spin–orbit coupling selects or mixes the pairing. It does not, at any level that matters here. The gallery state is sodium-derived, with only a few percent cobalt weight, so its effective on-band spin–orbit scale is a couple of meV; the Rashba splitting from the off-centre sodium, estimated from the displaced charge, is about 0.8 meV, of order the gap but far below the Fermi energy, and the parity-mixed component of the gap it induces is at the sub-percent level. On the average centrosymmetric structure the up and down sectors carry opposite Rashba fields and largely cancel. Spin–orbit coupling is a small correction to the spin response, not a term in the pairing.

d. A probe-dependent valence. If the sodium keeps a fraction q of its electron in the gallery rather than giving it to the cobalt, then a site-specific valence probe—the cobalt L-edge or NQR—and a charge-counting probe—redox titration of the whole crystal—should disagree, by $\Delta v = x\,q \approx 0.04$ per cobalt, and only when the gallery is open. We are careful here: this small split is not the large valence offset seen in these materials [8], which needs an extra electron reservoir such as oxonium and is about six times larger; our prediction is the small, gallery-specific piece that switches on with the water and vanishes in the monolayer hydrate where the gallery band is empty.

e. Predictions for related systems. The picture points beyond this one compound, and we label each as a prediction. A gated $LiCoO_2$ thin film, in which the carrier density is set electrostatically rather than chemically, should show the interlayer gas turn-on without the

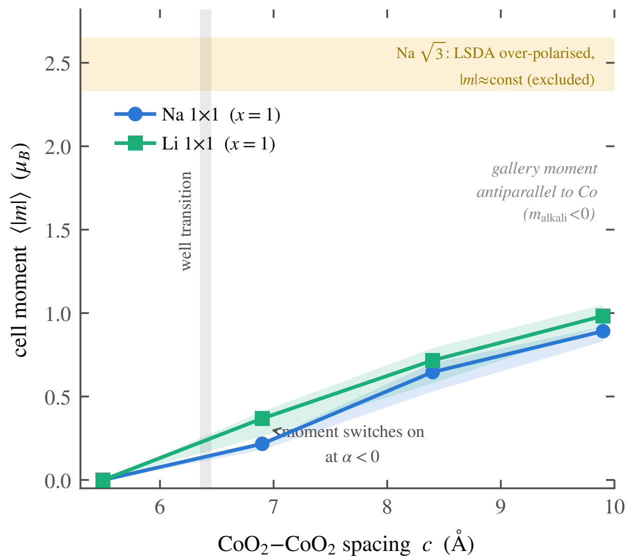

FIG. 5. The magnetic moment turning on with the well transition. Cell-integrated moment $\langle|m|\rangle$ for Na 1×1 and Li 1×1 against gallery spacing; markers are the mean over the displacement scan and the bands its spread. The local moment switches on at the same spacing where α turns negative, and the gallery moment is antiparallel to the cobalt. The $\sqrt{3}$ cell is over-polarized in LSDA and is shown only as an excluded caveat band.

disadvantage of the light frozen ion—a place to test the gas and the magnetism apart from the glue. Potassium cobaltate K_xCoO_2, with a still larger and heavier gallery ion, should reach the soft-mode window at a wider spacing and, if it can be hydrated to open the gallery, should superconduct by the same route with a heavier, softer mode. The mixed-anion cobaltate Na_2CoSe_2O [9] carries the same two-dimensional cobalt network with a different gallery chemistry, and our picture asks a definite question of it: does its gallery ion sit in a soft anharmonic well, and does a gas condense there? Finally the layered nitride superconductor Li_xZrNCl deserves re-analysis in the same language, as a gallery whose carrier density and interlayer mode may be doing together what they do here. Each is one experiment, and each is stated so it can fail.

VI. INADEQUACIES OF THE CALCULATION

It is worth setting down plainly what this calculation cannot do, because the softest numbers are load-bearing and the reader should know which they are.

The largest systematic is that we held the oxygen planes fixed at one height for every spacing and displacement. Moving them by 0.06 Å shifts single-point energies by as much as the well depth itself, so every well depth here carries a systematic of about $\pm50\%$. This does not touch the sign of α: the single-to-double-well flip survives $\pm50\%$ scaling and survives the convergence checks, and the polaronic coupling in the frozen regime is an order of magnitude past the limit, so no conclusion turns on the soft depths. But the peak height of the dome does depend on them, and we do not claim 14 K to better than the factor of two already stated.

The rest of the list is shorter. Our transition spacing is computed at full sodium coverage, $x = 1$, and only checked at the physical $x = 1/3$ in a single cell, where it moves by about 0.2 Å. The local-spin-density treatment over-polarizes the $\sqrt{3}$ cell, giving an unphysically large and flat moment there; we therefore rest the magnetic story on the 1×1 series and exclude the $\sqrt{3}$ moments, and we have checked that the well energies themselves are insensitive to the spin treatment at the few-percent level, so the over-polarization does not touch the wells. There is no explicit water anywhere in these calculations, which is exactly the gap that Sec. IV says must be filled next. And the pairing is treated as a single Einstein boson in the Allen–Dynes approximation, with an explicit cut where the coupling leaves the regime that approximation can describe; a real bipolaron treatment past that cut is not attempted. The final test of any physical theory lies, of course, in experiment, and Sec. IV and Sec. V give the ranked list, the deuteration shift and the soft-mode search first.

TODO — PENDING CALCULATIONS, TO BE INTEGRATED BEFORE SUBMISSION

- **TODO — Explicit-water DFT (Sec. IV)** — $E(\delta)$ for Na in the bilayer-hydrate geometry with four solvating H_2O vs. the same cell with vacuum; the water-softening claim stands or falls here. Batch built (set G), awaiting the GPU run.
- **TODO — D_2O isotope shift (Sec. IV)** — from the water-softened well, re-quantize with D_2O masses; predicted T_c shift against the existing deuteration experiments.
- **TODO — $Li_xZrNCl/HfNCl$ reanalysis (Sec. V)** — published T_c vs. basal-spacing data confronted with the gallery mechanism; support, refutation, or orthogonal. Literature extraction in progress.
- **TODO — K_xCoO_2 scans (Sec. V)** — set E; does the larger alkali widen the superconducting window?
- **TODO — Gated thin-film $LiCoO_2$ (Sec. V)** — set F, jellium-gated monolayer at 5.5 Å; do added carriers occupy the gallery band while Li stays quantum-paraelectric? The ionic-liquid-gating prediction for the thin-film collaboration.
- **TODO — Na_2CoSe_2O gallery states (Sec. V)** — set H; does the stable Na_2O gallery of the 2024

EXPERIMENTAL FACT	OUR VERDICT	WHY IT FOLLOWS
Anhydrous (5.5 Å) is not SC *Foo 2003*	explained	$\alpha > 0$: single stiff well ($\hbar\omega \approx 30$ meV), $N(0)\approx 0$ — no 2DEG, nothing to pair.
Monolayer hydrate (6.9 Å) is not SC *Foo 2003*	explained	Past the window: $\lambda \approx 43 \gg 2$ — the alkali self-traps (polaronic), so SC is killed.
Bilayer hydrate (9.9 Å) is SC, T_c=4.5 K *Takada 2003*	prediction	Bare geometry is polaronic here; SC needs water to soften the Na well back into the window (untested).
No superconducting Li analogue	explained	$c^*_{\mathrm{Li}} \lesssim 5.5$ Å and lighter mass $\Rightarrow$ stiffer mode; Li never enters the soft-mode window.
Magnetic phase borders the dome *expt.*	explained	Local moment turns on exactly at the $\alpha < 0$ well transition (Fig. 4), coincident with the dome.
Superconducting T_c is only a few K *Takada 2003*	explained	Allen–Dynes with DFT ω_{eff}, λ gives peak $T_c \approx 14$ K (band 11–23) — the right scale.

Foo *et al.*, Solid State Commun. **127**, 33 (2003) · Takada *et al.*, Nature **422**, 53 (2003)

FIG. 6. The mechanism against the record. Six established facts about $Na_xCoO_2 \cdot yH_2O$ and what the soft-mode, charge-bifurcation picture makes of each: five are explained by the static calculation, and one—the superconductivity of the bilayer hydrate at 9.9 Å—is a prediction contingent on water re-softening the sodium well (Sec. IV). The one-line reason for each verdict is given.

$T_c = 6.3$ K compound host the same interlayer band?

- **TODO — Site-specific XAS/NQR numbers (Sec. V)** — the probe-dependent Co-valence split $x\,q$ evaluated from the v4 Bader charges, with the measurable magnitude stated.

- **TODO — Design-rule novelty audit** — prior-art verdict ($SrTiO_3$ quantum criticality, C_6Ca interlayer state, KOs_2O_6 rattling) and the one-sentence claim we can defend; in progress.

VII. SUMMARY

Sodium cobaltate superconducts only when it is wet. We have shown, by ordinary density-functional calculations, that opening the gallery between its CoO_2 sheets past a critical spacing does two things at once—condenses an interlayer electron gas and softens the sodium into a rattling double well—and that a phonon-mediated dome lives in the narrow window between the layers too close and the sodium frozen. The dry parent and the monolayer hydrate fall correctly outside that window. The one compound that superconducts sits, in the static calculation, on the wrong side of it, and that mismatch is the paper's real lesson: the water is not the spacer that holds the gallery open but the lubricant that keeps the sodium soft at a spacing where a rigid lattice would freeze it. The mechanism is testable, and the test that decides it is a calculation of the sodium well with the water put in and allowed to move.

ACKNOWLEDGMENTS

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